

Predictive Maintenance of Industrial Machinery

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Project Link- [Click_Here](#) (github)

Program	AICTE IBM SkillsBuild Summer Internship Program (SIP) 2026
Organised by	Edunet Foundation in collaboration with IBM
PS Number	Problem Statement No. 39
Category	Machine Learning (ML) using AI_KOSH / Kaggle Datasets
Branch	Mechanical Engineering — applicable to CS / IT students
Dataset	Machine Predictive Maintenance Classification — Kaggle
Dataset Link	kaggle.com/datasets/shivamb/machine-predictive-maintenance-classification
Technology	IBM Cloud Lite services — IBM watsonx.ai Studio (AutoAI)

Problem Statement

Industrial machines such as lathes, CNC mills, pumps, and hydraulic systems are prone to unexpected failures that cause significant production loss, costly downtime, and high repair expenses. Traditional maintenance strategies are either **reactive** (repairing after failure occurs) or **scheduled** (time-based service regardless of machine condition) — both of which are inefficient, wasteful, and fail to prevent unplanned breakdowns.

There is a critical industry need for a **data-driven predictive system** that can detect failure patterns early using real-time sensor data from machines. Such a system would enable maintenance teams to act proactively — replacing components just before failure rather than too early or too late.

TASK

Design, build, and evaluate a **multi-class classification model** that can accurately predict the type of machine failure based on sensor readings. The model must classify each machine state into one of the following five categories:

- **No Failure:** Machine is operating within normal parameters — no action required.
- **Tool Wear Failure:** Tool wear exceeds safe threshold (typically >200 min) — tool replacement needed.

- **Heat Dissipation Failure:** Air/process temperature difference too low with low RPM — cooling issue detected.
- **Power Failure:** Extremely high RPM with very low torque — power system anomaly.

Dataset Details

Feature	Type	Description
Type	String	Machine quality variant — L (Low), M (Medium), H (High)
Air temperature [K]	Decimal	Ambient air temperature in Kelvin
Process temperature [K]	Decimal	Process/operating temperature in Kelvin
Rotational speed [rpm]	Integer	Machine spindle rotational speed in RPM
Torque [Nm]	Decimal	Torque applied to the machine in Newton-metres
Tool wear [min]	Integer	Cumulative tool wear time in minutes
Target (binary)	Integer	Binary: 0 = No failure, 1 = Failure occurred
Failure Type	String	TARGET — Multi-class failure category to predict

■ Dataset size: ~10,000 rows | Target column: Failure Type | Format: CSV | Source: Kaggle (shivamb/machine-predictive-maintenance-classification)

Mandatory Technology

- **IBM Cloud Lite Account** — Free tier account required at cloud.ibm.com
- **IBM watsonx.ai Studio** — Project workspace for building AutoAI experiments
- **IBM watsonx Machine Learning** (WatsonMachineLearning service) — For model training, evaluation, and deployment
- **AutoAI** — Automated ML pipeline generation, feature engineering, and hyperparameter optimization
- **IBM Granite Models** — Mandatory IBM AI model family requirement per program guidelines

Evaluation Criteria

- Model **Accuracy** on test/hold-out data (primary metric)
 - **Precision, Recall, and F1 Score** per failure class
 - Quality of **Feature Engineering** and preprocessing steps
 - Correct use of **IBM watsonx.ai AutoAI** platform
 - Clarity of **problem understanding** and solution documentation
 - **GitHub repository** with all required files uploaded correctly
 - Quality of **PowerPoint presentation** and panel viva performance
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